

Harmonicity accounts for the tritone’s shortfall on the dissonance curve for dyads

Daniel Chen

Independent Researcher, Taipei, Taiwan

boenchen1112@gmail.com ORCID [0009-0009-5840-5012](https://orcid.org/0009-0009-5840-5012)

Abstract

Sensory dissonance curves rate the tritone about as rough as the minor sixth, yet listeners prefer the minor sixth. We reanalyse 95,865 pleasantness ratings of dyads sampled continuously from 0 to 15 semitones, in ten US and three Korean experiments that vary the tone spectrum. Over six harmonic spectra, US listeners rate the tritone +0.26 within-listener SD below the minor sixth, and a roughness curve calibrated on all other intervals reproduces none of this gap. The primary estimand, the gap left on held-out experiments by a curve that adds harmonicity with a wider tolerance of mistuning, is +0.18 SD [+0.04, +0.32] (random effects, Hartung–Knapp), with a prediction interval for a new spectrum of [−0.18, +0.55]. This remainder shows no detectable tritone penalty. After harmonicity the tritone’s own residual is −0.02 [−0.15, +0.11], and the gap is the minor sixth rated above the curve (+0.17 [+0.03, +0.31]), as the other consonant intervals are (difference from their mean +0.02 [−0.15, +0.18]). An exploratory bonus for in-tune intervals, and a step in the curve’s error, differ between spectra. With stretched partials the tritone dip follows the stretched scale; the in-tune bonus does not. None of four targeted tests (culture, training, third harmonic, roll-off) survives Holm correction. The curves’ largest failure is elsewhere: they capture 59% of the reliable variance, missing mainly a smooth trend over interval size. For dyads, once harmonicity is in the curve, no penalty specific to the tritone is detectable.

Keywords: consonance; dissonance curve; tritone; roughness; harmonicity; equal temperament; noise ceiling; Hutchinson–Knopoff; cross-cultural; timbre

1 Introduction

A sensory dissonance curve sums, over every pair of partials in a sound, the roughness that two pure tones produce when their separation falls within a fraction of a critical bandwidth (Plomp & Levelt, 1965; Kameoka & Kuriyagawa, 1969; Hutchinson & Knopoff, 1978; Sethares, 1993). For harmonic complex tones such curves reproduce much of the familiar ranking of intervals, with minima at the octave, fifth and fourth and maxima near the semitone and major seventh (Sethares, 2005), and they are often presented as an explanation of consonance.

The tritone is an awkward case. In Western theory and pedagogy it is the dissonance *par excellence*; on a roughness curve it sits at about the same height as the minor sixth (Figure 1). The usual reply is that the curve is an oversimplification and that factors it does not capture decide the rest (Parncutt & Hair, 2011; Harrison & Pearce, 2020; Lahdelma et al., 2022). That reply is plausible but not quantitative: it does not say how large the discrepancy is, whether a better curve removes it, or whether what remains belongs to the tritone at all.

The dense dyad ratings of Marjieh et al. (2024) make these questions answerable. Their intervals were sampled continuously from 0 to 15 semitones, for many tone spectra and for

listeners in the United States and South Korea. Continuous sampling lets the rating at the tritone be compared with a curve fitted to all other intervals, so the discrepancy can be estimated without trusting the model’s absolute scale. Varying the spectrum asks whether the discrepancy follows the partials, as a psychoacoustic mechanism would, or stays with the interval, as a category attached to the frequency ratio would.

We make five claims. (i) In isolated dyads the tritone is not the least pleasant interval, but listeners rate it below the minor sixth, and a roughness curve calibrated on all other intervals reproduces none of that gap. (ii) Adding harmonicity, the degree to which the combined spectrum resembles a single harmonic series (Terhardt, 1974; Parncutt, 1989; Harrison & Pearce, 2020; Bowling & Purves, 2015), removes the tritone’s own shortfall: its residual is no longer distinguishable from zero or from that of the other intervals. A gap remains, but it is the minor sixth rated above the curve, as the other consonances are; it varies between spectra and is unexplained. (iii) Of the stimulus-based factors fixed before a held-out screen, only a wider tolerance of mistuning in the harmonicity model improves prediction of unseen experiments. The resulting curve is the one we recommend; it is a small step beyond the published model. (iv) With stretched partials the tritone dip leaves 600 cents and follows the stretched scale, whereas the bonus that in-tune intervals of harmonic tones receive does not; none of four targeted tests of what the residual depends on survives correction for multiplicity. (v) The tritone is not where the curves fail most: against the reliability of the ratings they capture about 59% of the explainable variance, and the largest missing component is a smooth trend over interval size. We also note that analyses of the chords of Bowling et al. (2018) based on `incon`, including an earlier analysis by the author, scored them in equal temperament although they were presented in just intonation (Section 5.7).

Relation to previous work. That consonance has more than one acoustic component is not new. Parncutt (1989) and Parncutt and Hair (2011) attribute the tritone’s dissonance to its being neither smooth nor harmonic, and Parncutt et al. (2019) note that roughness alone cannot order the fourth and the minor sixth. Harrison and Pearce (2020) model chord consonance as interference, periodicity/harmonicity and cultural familiarity, Eerola and Lahdelma (2021) compare acoustic and cultural predictors across chord data sets, Marjeh et al. (2024) fit a composite of revised roughness and harmonicity to the dense dyad data we reuse, and Masina et al. (2022) combine roughness with a periodicity term for dyads. Ratio-based measures, from Euler’s gradus to a recent asymmetric variant (De Roure, 2026), score only exact ratios and so make no prediction along a continuous interval axis. Marjeh et al.’s code defines, but does not use, a harmonicity variant with a wider template (Harrison, 2023), so the idea of a wider tolerance is not ours either. What is new here is a tritone-specific estimate of the discrepancy, calibrated on all other intervals and tested on held-out experiments; its location at the minor sixth rather than the tritone; a stretched-spectrum test that separates the dip from the in-tune bonus; and a noise ceiling showing that the curves’ largest failure is not local at all. The two readings of the residual we discuss, an incomplete harmonicity model or a learned association, map onto the current debate between psychoacoustic accounts (Harrison & Pearce, 2020; Marjeh et al., 2024) and the view that harmonicity partly stands in for cultural familiarity (Lahdelma et al., 2022; McDermott et al., 2016; Milne et al., 2023).

2 Data

Dense dyad ratings. Marjeh et al. (2024) asked online participants to rate the pleasantness of two-tone chords on a seven-point scale. The interval was drawn from a continuous uniform distribution between 0 and 15 semitones, so each listener heard a different random set of intervals. Ratings and the synthesis parameters of every timbre were obtained from the authors’ repository (Harrison, 2023). We used every dyad experiment with a fixed timbre and US participants (Table 2): harmonic complex tones with a 3 dB/octave roll-off; the same tones with partials stretched to an octave ratio of 2.1 or compressed to 1.9; five equal-amplitude harmonics with and without the third harmonic; pure tones; a bonang timbre whose lower tone is harmonic; and resynthesised flute, guitar and piano tones. For the flute, guitar and piano the spectrum available to us is a single set of harmonic amplitudes from one analysed tone, a proxy for note-specific samples. The Korean cohort rated the harmonic, stretched and compressed stimuli. A further experiment (Study 3 of Marjeh et al., 2024) varied the spectral roll-off continuously across the trials of the same 295 participants; we analyse it at three pre-set roll-off levels (2, 7 and 12 dB/octave, with a Gaussian kernel of SD 1.5 dB/octave over roll-off) and, for the exploratory analysis of Section 5.8, at five levels from 1 to 13 dB/octave. The 13 fixed-timbre experiments comprise 95,865 ratings from 1,627 US and 122 Korean participants (110,565 ratings with Study 3). Musicians are participants with more than two years of training and non-musicians those with two or fewer, the split used by Marjeh et al. (2024). Participant identifiers are local to each experiment, so we treat experiments as independent samples; any listener who took part in more than one would make the pooled intervals slightly too narrow.

Chord ratings. Bowling et al. (2018) collected consonance ratings for all 298 chords formed by adding one to three chromatic notes within an octave above a root, on a four-point scale, from 30 listeners. The stimuli were piano tones in just intonation: the tritone as 7/5 (582.5 cents) and the minor sixth as 8/5 (813.7 cents), as recorded in *inconData* (Harrison, 2019). Johnson-Laird et al. (2012) collected ratings for 103 triads and tetrachords in wide voicings, in equal temperament. Both sets were obtained from *Digital and Cognitive Musicology Lab*, EPFL (2025) and cross-checked against Harrison (2019).

3 Models

Roughness. The Hutchinson–Knopoff model (H&K) expresses the separation of two partials in critical bandwidths, $y = \Delta f / \text{CBW}(\bar{f})$ with $\text{CBW} = 1.72 \bar{f}^{0.65}$, and weights each pair by $g(y) = ((y/0.25)e^{1-y/0.25})^2$ for $y \leq 1.2$ (zero beyond) times the product of the amplitudes, normalised by $\sum a_i^2$ (Hutchinson & Knopoff, 1978). We also report the Sethares kernel (Sethares, 1993) and the revised H&K model of Marjeh et al. (2024), which adds a slow-beating term and an amplitude exponent fitted to their data.

Harmonicity. We implemented Harrison and Pearce’s harmonicity model (Harrison & Pearce, 2018, 2020). Each partial contributes a Gaussian (SD 6.83 cents, the published tolerance) to a 1200-bin pitch-class spectrum, which is cross-correlated with an 11-harmonic template at every rotation; harmonicity is the Kullback–Leibler divergence of the resulting virtual-pitch profile from a uniform distribution. For dyads we follow Marjeh et al. (2024) in using the stimuli’s partial amplitudes and template weights $1/i$; for chords we follow *incon* (Harrison, 2024), with idealised 11-harmonic tones and weights $i^{-0.75}$. With the lower tone at C4, our H&K, harmonicity and revised H&K profiles correlate with those published by Marjeh et al.

(2024) at $r \geq 0.9999$, 0.9996 and 0.9999 in all 13 experiments; on the Bowling chords our implementations reproduce `incon` exactly when given the same equal-tempered pitches.

Familiarity and periodicity. For the chords we also use Harrison and Pearce’s chord-frequency model, trained on popular-music transcriptions (Burgoyne et al., 2011), and Stolzenburg’s periodicity model (Stolzenburg, 2015), as computed by `incon`.

4 Estimands and inference

Profiles. Following Marjeh et al. (2024), ratings were z -scored within each participant and the pleasantness profile was estimated with a Gaussian kernel smoother (SD 0.2 semitones) on a 0.01-semitone grid. Model predictions for each trial were smoothed with the same kernel. All effects are in units of a listener’s own rating SD; for harmonic tones that SD is 1.24 points on the seven-point scale.

The gap, the dip and the residual. The *gap* is $P(8) - P(6)$, how much more pleasant the minor sixth is than the tritone; the *dip*, $\frac{1}{2}[P(5) + P(7)] - P(6)$, compares the tritone with the fourth and fifth. A model’s output was mapped onto the listener profile by least squares over 0.5–14.75 semitones, excluding windows of ± 0.6 semitones around the tritone and the minor sixth. The residual $r(c)$ is observed minus calibrated prediction at interval c ; the unexplained gap is $r(8) - r(6)$. Descriptive results (Table 2) calibrate each model on the experiment itself. Held-out results calibrate it with weights fitted to the other US experiments (leave one experiment out), with an intercept per experiment that absorbs its rating level and has no effect on the shape of the curve.

Primary estimand. The question of the paper, how much of the gap a dissonance curve leaves, was fixed before any analysis. For this version we declare one number as its answer: the held-out unexplained gap under the curve selected by the pre-specified screen of Section 5.4 (H&K roughness plus harmonicity with a 20-cent tolerance), pooled over the 6 US experiments with harmonic spectra. Every other estimate is secondary; those found after seeing the data are labelled exploratory where they appear.

Uncertainty and pooling. Confidence intervals come from 1,000 bootstrap resamples of participants, with the whole pipeline repeated in each. Experiments were combined by random-effects meta-analysis with the DerSimonian–Laird estimate of the between-experiment variance (DerSimonian & Laird, 1986) and a Hartung–Knapp interval, which uses a t distribution on $k - 1$ degrees of freedom and a weighted residual variance and is better calibrated when experiments are few (Hartung & Knapp, 2001; IntHout et al., 2014). We report τ , I^2 (Higgins & Thompson, 2002) and a 95% prediction interval for a new experiment. DerSimonian–Laird intervals are in Supplementary Table S1; of 326 pooled estimates, 22 change whether their interval excludes zero (listed in the released results). Four targeted tests (Section 5.6) form one pre-specified family, corrected by Holm’s procedure (Holm, 1979). The composite of Marjeh et al. (2024) was fitted to these same data, so its residuals are partly in-sample.

Every interval. To ask whether the tritone is unusual, the held-out residual was computed at every integer interval from 1 to 14 semitones, pooled over the harmonic spectra, and the tritone’s residual was compared with the mean of the other intervals from the minor second to

Table 1: Mean consonance ratings for the 12 dyads of [Bowling et al. \(2018\)](#) (higher = more consonant; the minor and major sevenths are tied).

Interval	Semitones	Rating	Rank (1 = least consonant)
Minor second	1	1.317	1
Minor seventh	10	1.753	2
Major seventh	11	1.753	3
Major second	2	1.837	4
Tritone	6	1.953	5
Minor sixth	8	2.477	6
Minor third	3	2.693	7
Major sixth	9	2.713	8
Perfect fourth	5	2.950	9
Major third	4	3.173	10
Perfect fifth	7	3.447	11
Octave	12	3.893	12

the major seventh. The minor sixth was likewise compared with the mean of the other consonant intervals (minor and major third, fourth, fifth, major sixth). Because the residual is relative to an intercept fitted on the rest of the interval axis, it shows where the shape of the curve fails, not how pleasant an interval is. The dip was also repeated at every integer interval, each with its own excluded window, as in an earlier analysis by the author.

Targeted tests. Four contrasts probe what the residual depends on: Korean minus US listeners for identical stimuli; musicians minus non-musicians; five harmonics without the third minus five harmonics with it, since the third harmonic creates the roughness that distinguishes the fourth and fifth from the tritone; and the change across spectral roll-off. Each is one pre-specified contrast on the residual after roughness and harmonicity.

Chords. For the chords we regressed ratings on chord-size indicators, the model outputs and the number of tritones in the chord; the coefficient of that count is the tritone penalty left unexplained (rating units, bootstrap CIs over chords). For the twelve Bowling dyads we report the share of the observed tritone–minor-sixth difference that each fitted model reproduces.

Noise ceiling. Each profile’s reliability was estimated by splitting its participants at random into halves (50 splits), smoothing each half, and applying the Spearman–Brown correction to the correlation between halves ([Spearman, 1910](#); [Brown, 1910](#)). The result is the R^2 a perfect model would reach against the observed profile.

5 Results

5.1 The premise

Among the twelve Bowling dyads (Table 1), 4 intervals are rated less consonant than the tritone (1.953), but the minor sixth is rated higher (2.477). The dense data show the same (Figure 1): the tritone sits in a local trough, well above the minor second and major seventh. That the tritone is “the most dissonant interval” describes its role in tonal syntax more than how an isolated dyad sounds ([Parncutt & Hair, 2011](#)). The narrower claim, that it is heard as less pleasant than a dissonance curve predicts, is the one we test.

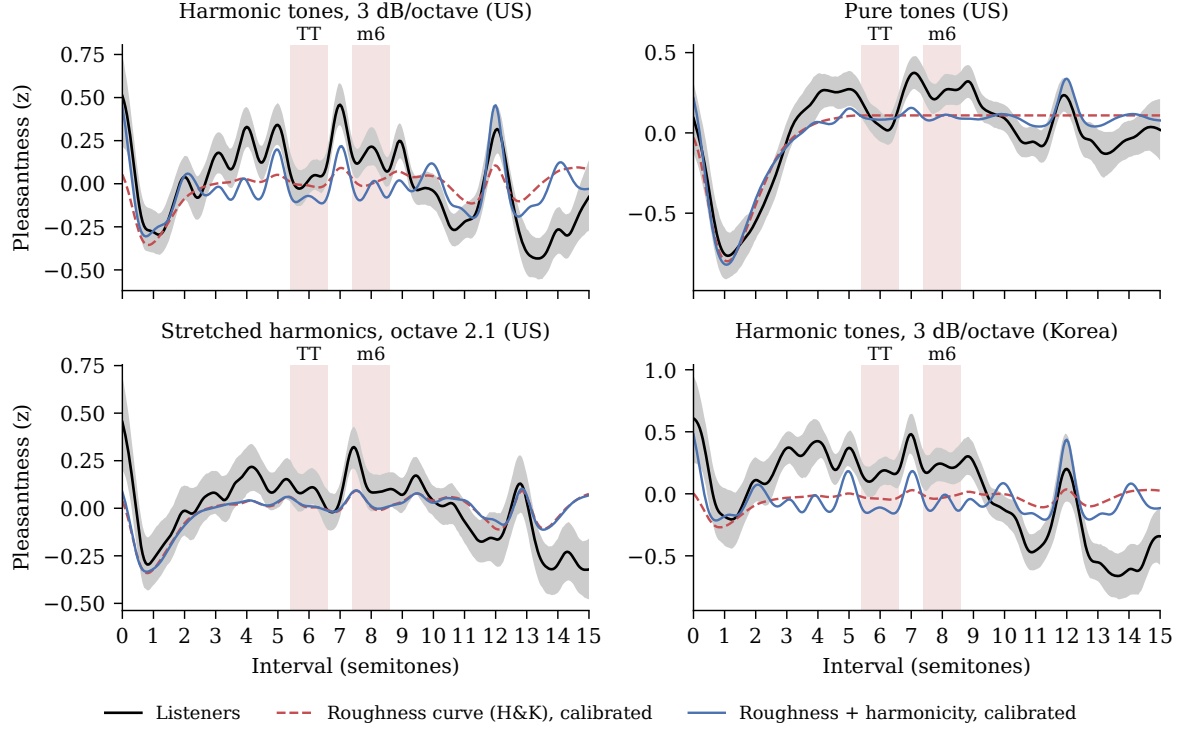


Figure 1: Listener pleasantness profiles (black; grey band, 95% bootstrap CI) with the roughness curve (H&K, red dashed) and the roughness-plus-harmonicity curve (blue), each calibrated to the listeners over all intervals outside the shaded windows at the tritone (TT) and minor sixth (m6). Harmonic tones (top left); pure tones (top right); stretched partials (bottom left); Korean listeners, harmonic tones (bottom right).

5.2 Roughness

Table 2 gives the gap in every experiment. For harmonic tones with a 3 dB/octave roll-off the minor sixth is rated $+0.19$ SD [$+0.04$, $+0.35$] above the tritone, and the gap left by calibrated H&K roughness is $+0.18$ SD [$+0.03$, $+0.33$]. Pooled over the six US experiments with harmonic spectra, the observed gap is $+0.26$ SD [$+0.16$, $+0.36$] and the gap left by calibrated H&K roughness is $+0.26$ SD [$+0.15$, $+0.37$], that is 100% of the observed gap (Table 4); the Sethares kernel leaves $+0.25$. Between the fourth and the fifth the roughness curve is nearly flat, so once calibrated it predicts almost the same pleasantness at the tritone and the minor sixth. Roughness separates the tritone from the fourth and fifth, whose low partials coincide, but at the tritone and the minor sixth the near-miss pairs of partials that do interact produce similar totals for these spectra. Its calibrated curve does reproduce the tritone’s dip below the fourth and fifth in part (pooled dip left $+0.25$, observed $+0.30$), which is why the dip and the gap tell different stories. For pure tones, whose partials do not interact at these intervals, roughness predicts no gap, yet listeners produce one ($+0.21$ SD [$+0.08$, $+0.34$]).

5.3 Harmonicity closes the tritone’s own shortfall

Adding harmonicity (published tolerance) leaves a pooled gap of $+0.19$ SD [$+0.02$, $+0.36$], 74% of the observed gap, with large heterogeneity ($I^2 = 83\%$; Table 4). The primary estimand, the held-out gap under the curve with a 20-cent tolerance, is $+0.18$ SD [$+0.04$, $+0.32$] ($\tau = 0.12$, $I^2 = 75\%$). Three similar values in this paper answer different questions: $+0.18$ is the primary estimand (held out, 20-cent tolerance, random effects with a Hartung–Knapp interval); $+0.19$ in

Table 2: The gap (minor sixth minus tritone), observed and left unexplained by each model calibrated on the experiment itself, in within-listener SD [95% participant-bootstrap CI]. Bold: CI excludes zero. The composite of Marjeh et al. (2024) was fitted to these data. N : participants.

Experiment	N	Observed	Left unexplained after			
			Roughness	Roughness + harmonicity	Composite	
<i>United States</i>						
Harmonic, 3 dB/oct	160	+0.19 [+0.04, +0.35]	+0.18 [+0.03, +0.33]	+0.12 [−0.03, +0.28]	+0.16 [+0.01, +0.31]	
5 equal harmonics	147	+0.18 [+0.07, +0.30]	+0.19 [+0.07, +0.31]	+0.04 [−0.08, +0.16]	+0.05 [−0.06, +0.16]	
5 harmonics, no 3rd	151	+0.16 [+0.06, +0.26]	+0.15 [+0.05, +0.25]	+0.01 [−0.10, +0.12]	0.00 [−0.10, +0.10]	
Flute	188	+0.35 [+0.22, +0.48]	+0.36 [+0.23, +0.50]	+0.34 [+0.20, +0.47]	+0.27 [+0.13, +0.40]	
Guitar	186	+0.39 [+0.25, +0.55]	+0.40 [+0.25, +0.55]	+0.37 [+0.23, +0.52]	+0.33 [+0.19, +0.48]	
Piano	191	+0.33 [+0.18, +0.48]	+0.32 [+0.18, +0.48]	+0.29 [+0.15, +0.45]	+0.25 [+0.11, +0.40]	
Pure tones	151	+0.21 [+0.08, +0.34]	+0.21 [+0.08, +0.34]	+0.18 [+0.05, +0.31]	+0.14 [+0.02, +0.26]	
Bonang (harmonic bass)	146	+0.38 [+0.22, +0.53]	+0.29 [+0.14, +0.44]	+0.31 [+0.16, +0.46]	+0.28 [+0.12, +0.42]	
Stretched (octave 2.1)	153	−0.02 [−0.17, +0.11]	−0.02 [−0.17, +0.11]	−0.02 [−0.17, +0.11]	−0.02 [−0.16, +0.12]	
Compressed (octave 1.9)	154	+0.02 [−0.12, +0.14]	−0.02 [−0.15, +0.10]	−0.02 [−0.15, +0.11]	−0.04 [−0.17, +0.08]	
<i>South Korea</i>						
Harmonic, 3 dB/oct	40	+0.07 [−0.12, +0.25]	+0.06 [−0.12, +0.24]	−0.03 [−0.20, +0.14]	+0.03 [−0.14, +0.21]	
Stretched (octave 2.1)	37	−0.17 [−0.34, 0.00]	−0.17 [−0.34, 0.00]	−0.18 [−0.34, 0.00]	−0.17 [−0.34, 0.00]	
Compressed (octave 1.9)	45	−0.21 [−0.39, −0.04]	−0.18 [−0.35, −0.02]	−0.14 [−0.31, +0.03]	−0.16 [−0.33, 0.00]	

Table 4 is the descriptive calibration, with the published tolerance and each experiment fitted to itself; and +0.19 in Section 5.4 is the held-out gap of the 20-cent curve as a plain mean over the six spectra. The primary estimand is a real gap, but the prediction interval for a new harmonic spectrum, [−0.18, +0.55], includes zero, and per spectrum it ranges from +0.03 (five harmonics without the third) to +0.36 (guitar). The synthetic spectra leave little (harmonic tones +0.12, five equal harmonics +0.07); the resynthesised flute, guitar and piano leave most (+0.29, +0.36 and +0.27). Because the spectra we have for those three instruments are proxies, this contrast may reflect spectral detail that the models do not see as much as a property of the listeners.

The descriptive calibrations agree. Harmonicity without roughness leaves less of the gap (+0.11) but fits the rest of the interval axis worse: outside the excluded windows it explains $R^2 = 0.41, 0.33$ and 0.21 of the profile for five equal harmonics, five harmonics without the third and the flute, against $0.65, 0.70$ and 0.75 for roughness plus harmonicity. The composite of Marjeh et al. (2024), whose parameters were fitted to these experiments, leaves +0.17, and harmonicity computed with *incon*’s template instead of Marjeh et al.’s leaves +0.20 [+0.05, +0.36]. The dip is more sensitive to how harmonicity is weighted: +0.30 SD observed, +0.14 [−0.02, +0.29] after roughness and harmonicity, +0.03 after harmonicity alone and +0.05 after the composite (Supplementary Table S2). Since the dip compares the tritone with its immediate neighbours, whereas the gap compares it with an interval on the other side of the fifth, the two need not agree, and the next paragraph shows why they do not.

Which end of the gap? A gap is a difference, $r(8) - r(6)$, so it can arise at either end. Table 3 splits it. Under roughness alone the tritone lies −0.09 [−0.22, +0.03] SD from its prediction and the minor sixth +0.17 [+0.06, +0.29]; relative to the mean residual of the other intervals from the minor second to the major seventh, the tritone is −0.18 [−0.30, −0.05] below. The tritone’s shortfall is therefore relative, not reliably negative in absolute terms, and it is clearest for the guitar, piano and five harmonics without the third. Adding harmonicity with the 20-cent tolerance moves the tritone to −0.02 [−0.15, +0.11] and its difference from the other intervals to −0.11 [−0.25, +0.02], neither distinguishable from zero, while the minor sixth stays at +0.17 [+0.03, +0.31]. What is left of the gap is the minor sixth rated above the curve. The minor sixth is not unusual in this: its residual does not differ from the mean of the minor and major thirds, fourth, fifth and major sixth (+0.02 [−0.15, +0.18]; Table 7). These residuals are

Table 3: Which end of the gap. Held-out residuals (observed – predicted) at the tritone, $r(6)$, and the minor sixth, $r(8)$, under H&K roughness and under the recommended curve (H&K + harmonicity, 20-cent tolerance), and the gap left by the latter, in SD [participant-bootstrap CI]. Pooled row: random effects over the six harmonic spectra, Hartung–Knapp CI; its last cell is the primary estimand. Bold: CI excludes zero.

Experiment	Roughness (H&K)		H&K + harmonicity (20 cents)		Gap left
	$r(6)$	$r(8)$	$r(6)$	$r(8)$	
Harmonic, 3 dB/oct	+0.03 [−0.07, +0.12]	+0.21 [−0.09, +0.32]	+0.10 [0.00, +0.19]	+0.22 [−0.10, +0.33]	+0.12 [−0.03, +0.27]
5 equal harmonics	+0.07 [−0.03, +0.17]	+0.25 [−0.18, +0.33]	+0.15 [−0.04, +0.25]	+0.22 [−0.14, +0.29]	+0.07 [−0.03, +0.19]
5 harmonics, no 3rd	−0.16 [−0.24, −0.08]	−0.02 [−0.10, +0.07]	−0.11 [−0.19, −0.02]	−0.08 [−0.17, 0.00]	+0.03 [−0.07, +0.13]
Flute	−0.07 [−0.18, +0.03]	+0.29 [−0.18, +0.39]	+0.02 [−0.09, +0.12]	+0.31 [−0.20, +0.40]	+0.29 [−0.15, +0.42]
Guitar	−0.23 [−0.34, −0.12]	+0.16 [−0.04, +0.29]	−0.16 [−0.27, −0.04]	+0.20 [−0.08, +0.32]	+0.36 [−0.20, +0.52]
Piano	−0.18 [−0.28, −0.09]	+0.14 [−0.04, +0.26]	−0.10 [−0.20, 0.00]	+0.17 [−0.06, +0.28]	+0.27 [−0.12, +0.42]
Pure tones	−0.06 [−0.16, +0.03]	+0.15 [−0.04, +0.27]	−0.01 [−0.10, +0.08]	+0.18 [−0.07, +0.30]	+0.19 [−0.06, +0.32]
Pooled, harmonic (RE, $k = 6$)	−0.09 [−0.22, +0.03]	+0.17 [−0.06, +0.29]	−0.02 [−0.15, +0.11]	+0.17 [−0.03, +0.31]	+0.18 [−0.04, +0.32]

Table 4: Random-effects pooled estimates [95% Hartung–Knapp CI] and 95% prediction intervals (PI) for a new experiment, SD units, with each model calibrated on the experiment itself. Harmonic: the six US experiments with harmonic spectra. Eight: adds pure tones and bonang. ^aFitted to these data. DerSimonian–Laird intervals: Supplementary Table S1.

	Gap, harmonic ($k = 6$)		Gap, eight ($k = 8$)		Dip, harmonic ($k = 6$)	
	Estimate [CI]	PI	Estimate [CI]	PI	Estimate [CI]	PI
Observed	+0.26 [+0.16, +0.36]	[+0.03, +0.49]	+0.27 [+0.18, +0.35]	[+0.08, +0.45]	+0.30 [+0.22, +0.37]	[+0.17, +0.43]
After roughness (H&K)	+0.26 [+0.15, +0.37]	[−0.01, +0.53]	+0.26 [+0.18, +0.33]	[+0.07, +0.44]	+0.25 [+0.13, +0.36]	[−0.03, +0.52]
After roughness (Sethares)	+0.25 [+0.13, +0.38]	[−0.05, +0.56]	+0.25 [+0.16, +0.34]	[+0.01, +0.49]	+0.28 [+0.17, +0.39]	[+0.03, +0.53]
After harmonicity alone	+0.11 [−0.05, +0.27]	[−0.32, +0.55]	+0.14 [+0.02, +0.27]	[−0.22, +0.51]	+0.03 [−0.10, +0.16]	[−0.31, +0.37]
After roughness + harmonicity	+0.19 [+0.02, +0.36]	[−0.26, +0.64]	+0.20 [+0.08, +0.32]	[−0.14, +0.55]	+0.14 [−0.02, +0.29]	[−0.25, +0.53]
harmonicity template $i^{-0.75}$ (incon)	+0.20 [+0.05, +0.36]	[−0.21, +0.61]	+0.21 [+0.10, +0.32]	[−0.10, +0.52]	+0.15 [0.00, +0.30]	[−0.23, +0.53]
After revised H&K + harmonicity	+0.28 [+0.11, +0.44]	[−0.16, +0.71]	+0.27 [+0.15, +0.38]	[−0.05, +0.58]	+0.20 [+0.02, +0.39]	[−0.28, +0.68]
After Marjeh et al. composite ^a	+0.17 [+0.03, +0.31]	[−0.20, +0.54]	+0.18 [+0.08, +0.28]	[−0.11, +0.46]	+0.05 [−0.13, +0.23]	[−0.43, +0.54]

relative to an intercept fitted on the rest of the interval axis, so they locate where the shape of the curve is wrong, not absolute levels. The same holds for pure tones (tritone −0.01, minor sixth +0.18; point estimates).

Tuning. Located with a narrower kernel (SD 0.1 semitones; exploratory), the listeners’ minor-sixth peak lies at 808 [796, 819] cents, between equal temperament and 8/5 (813.7 cents) and −6 [−17, +6] cents from the harmonicity model’s peak, and the gap left is no smaller at just-intoned positions (difference +0.03 [−0.11, +0.17]), so these data cannot tell whether listeners heard the minor sixth as tempered or just (inconclusive; Supplementary Section S9).

5.4 A better curve on held-out spectra

With one pair of weights fitted across US experiments (published tolerance) and each experiment predicted from weights fitted to the others (Table 5), the held-out unexplained gap is positive in 8 of 10 experiments and its CI excludes zero in 5; for harmonic spectra, in 6 and 3 of 6. The residual on unseen spectra is in the same range as the descriptive one, so it is not an artefact of refitting each experiment.

A pre-specified screen. Before looking, we fixed a list of candidate third factors, all computed from the stimulus and none tied to a particular interval, and one criterion: prediction of held-out experiments. The candidates were (1) a wider tolerance of mistuning in the harmonicity model, the Gaussian width σ of spectrum and template chosen from 6.83, 10, 15 or 20 cents inside each fold; (2) root ambiguity (Terhardt, 1974; Parncutt, 1988), the height of the second-best peak of the virtual-pitch profile relative to the best; (3) temporal periodicity, the largest normalised autocorrelation of the partials over lags of 2–30 ms (Pressnitzer et al., 2001),

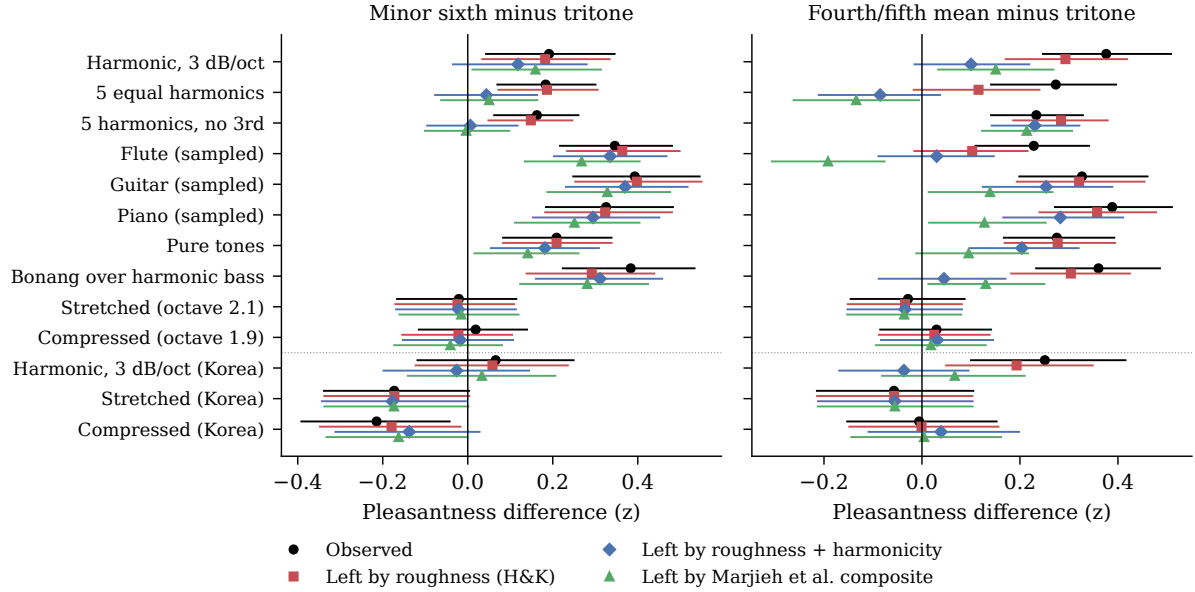


Figure 2: Observed gap and dip (black) and the part left unexplained by roughness (red), roughness plus harmonicity (blue) and the fitted composite of Marjeh et al. (2024) (green), per experiment, each model calibrated on the experiment itself, with 95% bootstrap CIs. Below the dotted line: stretched and compressed spectra and the Korean cohort.

in the spirit of Stolzenburg (2015) and Masina et al. (2022); and (4), as a deliberate foil, the harmonic entropy of the frequency ratio (Tonalsoft, n.d.), which ignores the spectrum. Baselines were refitted inside each fold. The list was fixed in advance but not publicly registered.

Only the tolerance helped (Table 6). The fold-wise choice was $\sigma = 20$ cents in every fold, and the two-term curve then reaches a mean held-out R^2 of 0.548, against 0.533 with the published tolerance (paired difference +0.015 [+0.005, +0.023]), 0.501 for revised roughness plus harmonicity and 0.493 for the composite of Marjeh et al. (2024) (paired difference +0.055 [+0.032, +0.079]; 500 participant bootstraps of the whole leave-one-out procedure). The composite’s internal weights were fitted to these experiments, so the comparison favours it. Root ambiguity ($R^2 = 0.527$) and periodicity ($R^2 = 0.517$) added nothing once harmonicity was present, and the foil added nothing ($R^2 = 0.519$). Checked afterwards (exploratory), the optimum depends on smoothing: with a kernel of 0.1 semitones the best tolerance is 10 cents, with 10, 15 and 20 cents nearly tied (0.548, 0.548, 0.546), and with 0.3 semitones it is 20. A tolerance wider than the published one helps at every width (gains 0.008, 0.015 and 0.018 in R^2), but 20 cents is a representative value, not an estimate. Rows examined after the screen are in Supplementary Table S3.

Before running the smoothing check we had set the criterion that the best tolerance should lie within 15–25 cents at both the narrower and the wider kernel; it was met at 0.3 but not at 0.1 semitones, where some folds chose 10 cents. The lower tone of each dyad was sampled from G3 to F4, whereas the models place it at C4; averaging H&K roughness over that range changes the held-out R^2 of the recommended curve by -0.001 [-0.001 , 0.000] and harmonicity is unaffected by transposition. Register does shift the overall liking of pure-tone dyads (Kaklamani & Simserides, 2026), but a shift common to all intervals of an experiment is absorbed by its intercept.

The tolerance improves the curve as a whole but barely moves the gap: the held-out gap in the six harmonic experiments is +0.19SD, against +0.20 with the published tolerance, and

Table 5: Leave-one-experiment-out prediction (published tolerance). b_R , b_H : roughness and harmonicity weights fitted to the other US experiments. Held-out unexplained gap and dip [95% CI], SD units. Bold: CI excludes zero.

Held-out experiment	b_R	b_H	Gap, roughness only	Gap, two mechanisms	Dip, two mechanisms
Harmonic, 3 dB/oct	-1.66	0.63	+0.18 [+0.03 , +0.33]	+0.15 [0.00, +0.30]	+0.15 [+0.02 , +0.29]
5 equal harmonics	-1.62	0.63	+0.19 [+0.07 , +0.30]	+0.09 [-0.02, +0.21]	-0.02 [-0.15, +0.11]
5 harmonics, no 3rd	-1.61	0.64	+0.15 [+0.05 , +0.25]	+0.04 [-0.06, +0.14]	+0.24 [+0.15 , +0.34]
Flute	-1.52	0.70	+0.36 [+0.23 , +0.49]	+0.30 [+0.17 , +0.44]	-0.01 [-0.13, +0.11]
Guitar	-1.54	0.70	+0.40 [+0.25 , +0.55]	+0.35 [+0.21 , +0.50]	+0.21 [+0.08 , +0.35]
Piano	-1.52	0.69	+0.32 [+0.18 , +0.48]	+0.28 [+0.14 , +0.44]	+0.25 [+0.14 , +0.38]
Pure tones	-1.55	0.71	+0.21 [+0.08 , +0.34]	+0.17 [+0.04 , +0.30]	+0.17 [+0.06 , +0.29]
Bonang (harmonic bass)	-1.68	0.66	+0.21 [+0.05 , +0.36]	+0.19 [+0.03 , +0.34]	+0.09 [-0.04, +0.22]
Stretched (octave 2.1)	-1.63	0.66	-0.03 [-0.17, +0.11]	-0.02 [-0.16, +0.11]	-0.04 [-0.15, +0.08]
Compressed (octave 1.9)	-1.66	0.65	-0.06 [-0.20, +0.06]	-0.06 [-0.19, +0.06]	+0.01 [-0.10, +0.12]

Table 6: Pre-specified out-of-sample screen (leave one US experiment out; point estimates). R^2 : mean over held-out experiments of the fit to the whole profile (10 US; six harmonic). Gap, dip: left unexplained, mean of the six harmonic experiments. TT – other: the tritone’s residual dip minus the mean of the other 13 intervals (eight unstretched experiments). Stretched: residual dip for stretched partials at 6.00 semitones and at the stretched scale’s tritone, 6.42. SD units. Post-hoc rows: Supplementary Table S3.

Model	Held-out R^2							Stretched, dip left	
	10 US	6 harm.	Gap left	Dip left	TT – other			at 6.00	at 6.42
<i>Existing models (weights refitted in each fold)</i>									
Roughness (H&K)	0.476	0.569	+0.27	+0.25	+0.26			-0.04	+0.08
Harmonicity alone	0.135	0.248	+0.14	+0.07	+0.05			-0.03	+0.18
H&K + harmonicity ($\sigma = 6.83$ cents)	0.533	0.626	+0.20	+0.14	+0.14			-0.04	+0.08
Revised H&K + harmonicity	0.501	0.620	+0.25	+0.15	+0.19			-0.04	-0.01
Composite of Marjeh et al.	0.493	0.558	+0.18	+0.06	+0.08			-0.04	+0.05
<i>Pre-specified candidates</i>									
H&K + harmonicity, tolerance σ chosen in fold	0.548	0.657	+0.19	+0.14	+0.15			-0.03	+0.09
H&K + harmonicity + root ambiguity	0.527	0.632	+0.21	+0.14	+0.15			-0.03	+0.08
H&K + harmonicity + periodicity	0.517	0.606	+0.20	+0.14	+0.14			-0.04	+0.08
H&K + periodicity	0.468	0.555	+0.26	+0.24	+0.25			-0.03	+0.08
H&K + harmonicity(σ) + ambiguity + periodicity	0.538	0.639	+0.17	+0.12	+0.13			-0.04	+0.09
H&K + harmonicity + harmonic entropy of f_0 ratio	0.519	0.630	+0.22	+0.14	+0.16			-0.08	+0.08

no candidate removed it. Tolerance, ambiguity and periodicity together reach $R^2 = 0.538$ and leave a gap of +0.17. The residual is concentrated in pure tones (+0.19) and the resynthesised flute, guitar and piano (+0.29, +0.36 and +0.27). The foil failed as intended on a second count: depending only on the frequency ratio, it necessarily puts its penalty at 600 cents for stretched tones as well (+0.05 SD in the fitted curve), where listeners show none (Section 5.6). With the dyads sampled continuously, a wider tolerance means only that listeners’ consonance peaks are broader than the model’s; it is not evidence about equal temperament as such (Section 5.7).

5.5 Is the tritone special?

Table 7 gives the held-out residual at every integer interval, pooled by random effects over the six harmonic spectra. Under roughness alone the most negative residuals between the minor second and the major seventh are those of the major second (-0.10) and the tritone (-0.09 [-0.22, +0.03]). After harmonicity with the 20-cent tolerance the consonant intervals, thirds, fourth, fifth and sixths, lie above the curve (the minor third +0.12, fourth +0.11, fifth +0.14, major sixth +0.19), while the seconds, the tritone and the minor seventh lie near it (tritone -0.02 [-0.15, +0.11], major second -0.09, minor seventh +0.03) and the major seventh lies above it (+0.09). The tritone is thus one of several intervals that the curve predicts about right, not

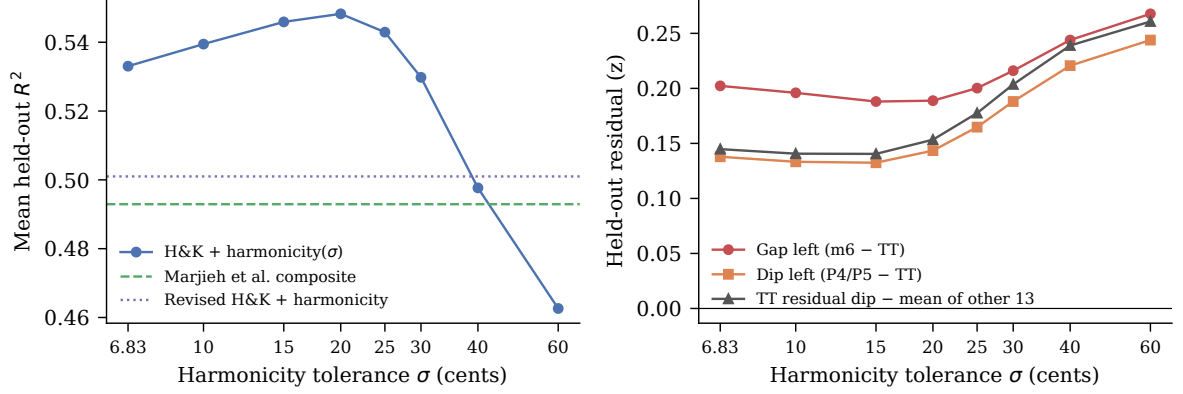


Figure 3: Harmonicity tolerance. Left: mean held-out R^2 of roughness plus harmonicity as the Gaussian width σ varies (published value 6.83 cents), against revised roughness plus harmonicity and the composite of Marjeh et al. (2024). Right: the tritone residuals left on held-out harmonic experiments. Values beyond 20 cents were examined after the screen.

Table 7: Held-out residual (observed – predicted) at each integer interval, random effects over the six harmonic US spectra, SD units. *Hartung–Knapp CI excludes zero. Size: a linear term in interval size (Section 5.8).

Model	m2	M2	m3	M3	P4	TT	P5	m6	M6	m7	M7	P8	m9	M9
H&K	-0.01	-0.10	+0.09	+0.19*	+0.14*	-0.09	+0.16*	+0.17*	+0.15*	0.00	+0.03	+0.25*	-0.12	-0.08
H&K + H (6.83)	-0.02	-0.14	+0.11	+0.18*	+0.07	-0.05	+0.10*	+0.16*	+0.18*	-0.02	+0.07	0.00	-0.08	-0.09
H&K + H (20)	+0.01	-0.09	+0.12*	+0.19*	+0.11	-0.02	+0.14*	+0.17*	+0.19*	+0.03	+0.09*	+0.04	-0.07	-0.05
H&K + H (20) + size	-0.01	-0.16*	+0.04	+0.11*	+0.04	-0.06	+0.09*	+0.15*	+0.18*	+0.05	+0.16*	+0.07	+0.04	+0.03

an interval detectably rated below it. Measured instead as a dip below its two neighbours, the tritone’s dip after harmonicity (published tolerance) differs from that of the average interval by $+0.16$ $[-0.01, +0.32]$ over the six harmonic spectra, an interval that includes zero, but by $+0.16$ $[+0.04, +0.27]$ when pure tones and the bonang are added; under the composite of Marjeh et al. (2024) the eight-spectrum figure is $+0.06$ $[-0.07, +0.20]$. These intervals are for the published tolerance; for the 20-cent curve only point estimates are available (Table 6). We report the eight-spectrum figure rather than drop it. Since the tritone’s own residual is not below average, that excess reflects the fourth and fifth rated above the curve on either side, not a shortfall at the tritone. Per-interval dips are shown in Supplementary Figure S1.

A bonus for in-tune intervals and a step (exploratory). Two further patterns were found after seeing the residual profile. Both are computed on the held-out residual of the curve with the 20-cent tolerance and a linear term in interval size (which absorbs the smooth trend of Section 5.8). The first is a local bonus: the residual at an equal-tempered interval c minus the mean at the quarter-tones on either side, $b(c) = r(c) - [r(c - 0.5) + r(c + 0.5)]/2$. Pooled over the harmonic spectra, the intervals from the minor second to the major seventh other than the tritone have a mean bonus of $+0.14$ $[+0.08, +0.19]$ SD and the tritone $+0.07$ $[0.00, +0.14]$; the individual bonuses run from $+0.04$ (minor seventh) and $+0.05$ (major second) to $+0.20$ (major third). The tritone’s shortfall relative to the other intervals, $+0.07$ $[-0.01, +0.14]$, includes zero. What the data do separate, and what we noticed only after seeing the profile, is that the major second and minor seventh receive a smaller bonus than the other eight intervals (-0.12 $[-0.14, -0.10]$, I^2 near zero); the tritone does not differ from them ($+0.03$ $[-0.04, +0.09]$). The minor second and the major seventh, which are rougher, receive the full bonus, and we do not know

Table 8: Decomposition of the held-out gap (exploratory; H&K + harmonicity at 20 cents + interval size), per harmonic spectrum [participant-bootstrap CI] and pooled by random effects [Hartung–Knapp CI]. Bonus part $b(8) - b(6)$; baseline part $q(8) - q(6)$; step: baseline rise between 7 and 8 semitones after a linear trend; other – TT: mean bonus of the other intervals minus the tritone’s. Bold: CI excludes zero. Guitar and piano use proxy spectra.

Spectrum	Gap left	Bonus part	Baseline part	Step	Other – TT bonus
Harmonic, 3 dB/oct	+0.14 [0.00, +0.29]	+0.13 [−0.02, +0.27]	+0.01 [−0.08, +0.11]	−0.02 [−0.12, +0.07]	+0.09 [−0.02, +0.19]
5 equal harmonics	+0.10 [−0.01, +0.22]	−0.02 [−0.15, +0.11]	+0.12 [+0.04, +0.20]	+0.04 [−0.03, +0.10]	−0.05 [−0.15, +0.04]
5 harmonics, no 3rd	+0.05 [−0.05, +0.15]	−0.02 [−0.12, +0.10]	+0.07 [−0.02, +0.16]	+0.09 [0.00, +0.17]	+0.12 [+0.04, +0.20]
Flute	+0.33 [+0.19, +0.47]	+0.16 [+0.02, +0.29]	+0.17 [+0.08, +0.26]	+0.01 [−0.08, +0.10]	+0.06 [−0.05, +0.16]
Guitar	+0.39 [+0.23, +0.55]	0.00 [−0.16, +0.17]	+0.39 [+0.28, +0.48]	+0.31 [+0.22, +0.41]	+0.04 [−0.08, +0.15]
Piano	+0.30 [+0.15, +0.45]	+0.16 [+0.03, +0.29]	+0.14 [+0.04, +0.24]	+0.17 [+0.08, +0.27]	+0.15 [+0.05, +0.25]
Pooled (RE, Hartung–Knapp)	+0.21 [+0.06, +0.36]	+0.06 [−0.03, +0.16]	+0.15 [+0.02, +0.28]	+0.10 [−0.03, +0.23]	+0.07 [−0.01, +0.14]
I^2 (%)	77	43	85	87	53

what separates the three intervals with a smaller one.

The bonus belongs to harmonic tones. Computed in the same way on the curve with a sharpness term (Section 5.8), it is +0.14 [+0.12, +0.15] for harmonic spectra but +0.04 [+0.01, +0.07] for pure tones and +0.04 [0.00, +0.07] for the bonang, and at either scale of the stretched and compressed tones it is not distinguishable from zero (Section 5.6). It survives changes of the harmonicity model: fitting two harmonicity terms of different width leaves most of it, and so do the published tolerance and Marjeh et al.’s revised roughness, which makes the slow beats of slightly mistuned intervals pleasant. At a smoothing width of 0.2 semitones the bonus cannot tell equal temperament from just intonation, so we read it as a bonus for the neighbourhood of the familiar intervals, not for their exact tuning. Marjeh et al. (2024) noted that details of the timbral effects remain unexplained by their models; we are not aware that this bonus has been quantified.

The second pattern is a step. Writing $r(c) = b(c) + q(c)$, with $q(c)$ the mean residual at the two neighbouring quarter-tones, the gap splits into a bonus part, $b(8) - b(6)$, of +0.06 [−0.03, +0.16], and a baseline part, $q(8) - q(6)$, of +0.15 [+0.02, +0.28]; the baseline rises between 7 and 8 semitones by +0.10 [−0.03, +0.23] as an indicator after a linear trend ($I^2 = 87\%$). On plain means over the spectra both parts exclude zero (+0.07 [+0.01, +0.13] and +0.15 [+0.11, +0.19]), but under random effects only the baseline part does, and the split is not stable across spectra (Table 8). The step is absent for the synthetic harmonic tone (−0.02 [−0.12, +0.07]) and the flute (+0.01 [−0.08, +0.10]), and large for the guitar (+0.31 [+0.22, +0.41]) and piano (+0.17 [+0.08, +0.27]). The guitar and piano spectra are proxies, one analysed tone each, so the step may come from the spectral model rather than from the listeners. The step is robust to the kernel width, the exclusion window and binning (Supplementary Table S5); what it is not robust to is the spectrum. We therefore do not decompose the gap into a fixed share of step and bonus.

5.6 What the residual depends on

Spectrum. With stretched or compressed partials the unexplained gap at 600 cents is about zero (US, after roughness and harmonicity: −0.02 and −0.02 SD), and so is the observed dip (−0.03 and +0.03), although the profiles are not flat (profile SD 0.15 and 0.13, against 0.22 for harmonic tones). A penalty attached to the physical 600-cent interval would survive stretching, since the fundamentals are still 600 cents apart. With partials stretched to an octave ratio r , the scale that matches the spectrum has its tritone at $6 \log_2 r$ semitones: 6.42 for the stretched and 5.56 for the compressed tones. There the stretched tones show an observed dip of +0.19 [+0.07, +0.31] SD, where there was none at 6 (paired difference +0.22 [+0.07, +0.37]). Held-out

H&K roughness predicts a dip of +0.12 at that position and leaves +0.08 $[-0.06, +0.20]$; the foil, attached to the frequency ratio, necessarily puts its penalty at 600 cents instead. For compressed tones the dip at 5.56 is not significant (+0.07 $[-0.05, +0.19]$), and the Korean listeners show no clear dip at either position, so the shift rests on one spectrum and one cohort.

The in-tune bonus of Section 5.5 does not follow the stretched scale. At the stretched scale’s own intervals the mean bonus is +0.02 $[-0.02, +0.05]$ (compressed +0.01 $[-0.02, +0.05]$), against +0.13 $[+0.10, +0.17]$ for harmonic tones, and the minimum effect these data could detect with 80% power is 0.05 SD (Supplementary Table S6; exploratory). The Korean listeners show the same pattern (harmonic +0.07 $[+0.03, +0.12]$, stretched $-0.04 [-0.08, +0.01]$). The dip moves with the partials; the bonus stays with harmonic tones.

The published harmonicity model cannot follow the shifted dip, because its template is a harmonic series. Replacing the template with the upper tone’s own partials, one reading of the proposal that harmonic templates are learned from the sounds heard (Shamma & Klein, 2000), reproduces the stretched-scale dip but predicts harmonic tones worse, so it is not a better general curve (exploratory; Supplementary Section S5).

Targeted tests. Four contrasts, fixed in advance, probe the residual after roughness and harmonicity (Table 9); the Holm correction over them was added in this version. Korean minus US listeners, same harmonic stimuli: -0.14 ($p = .227$). Musicians (more than two years of training) minus non-musicians, pooled over the eight unstretched US experiments: $+0.10$ ($p = .121$). Five harmonics without the third minus with it, observed change in the dip minus the change predicted by roughness ($-0.21 [-0.25, -0.17]$): $+0.17$ ($p = .045$). The change in the gap from 2 to 12 dB/octave roll-off, observed minus roughness-predicted: -0.04 ($p = .746$). After Holm correction 0 of the four remain significant (adjusted p from .180 to .746).

Taken individually, the contrasts say the following. Musicians show a larger observed gap and dip ($+0.16 [+0.06, +0.27]$ and $+0.26 [+0.16, +0.36]$; Table 10), part of which follows from harmonicity, to which musicians are more sensitive (McDermott et al., 2010). After roughness and harmonicity the gap is $+0.27 [+0.09, +0.44]$ for musicians and $+0.15 [+0.05, +0.25]$ for non-musicians, and the differences include zero (gap $+0.10 [-0.03, +0.23]$, dip $+0.12 [-0.01, +0.25]$), so the data do not establish a larger residual in trained listeners; the residual is present in non-musicians. For stretched and compressed partials the groups do not differ (gap difference $-0.10 [-0.29, +0.10]$; fixed-effect pooling over two experiments). Korean listeners show a smaller gap for harmonic tones, but they differ from US listeners in the same direction for stretched and compressed partials ($-0.15 [-0.36, +0.07]$ and $-0.23 [-0.45, -0.02]$), where US listeners show no gap, so the difference is not specific to consonance. Removing the third harmonic changes the observed dip less than roughness predicts (observed change $-0.04 [-0.20, +0.11]$), consistent with Marjeh et al. (2024), who found that the third harmonic matters through harmonicity; after correction this is suggestive only. In the roll-off experiment the gap is $+0.27 [+0.10, +0.44]$, $+0.18 [+0.05, +0.32]$ and $+0.29 [+0.16, +0.43]$ SD at 2, 7 and 12 dB/octave, while calibrated roughness predicts $+0.01 [+0.01, +0.02]$, $0.00 [-0.01, +0.01]$ and $0.00 [0.00, 0.00]$: the gap does not shrink as the upper partials, and with them the roughness, fade (Supplementary Table S7).

Culture and the smooth trend. The models fit the Korean profiles poorly (roughness plus harmonicity, in-sample $R^2 = 0.14$ for harmonic tones), although their ceilings are high (0.946–0.950). Predicted with weights from US listeners, the recommended curve does worse than each experiment’s mean (held-out $R^2 -0.19 [-0.26, -0.08]$ on average); adding a linear term in interval size brings it to $+0.11 [-0.01, +0.22]$ and a sharpness proxy to $+0.24 [+0.09, +0.36]$ (Supplementary Table S8; exploratory). What separates the cohorts in these data is thus

Table 9: The four pre-specified targeted tests, on the residual after roughness and harmonicity (published tolerance). SE from the participant bootstrap (musicianship: random effects over eight experiments, Hartung–Knapp SE, t on 7 df). p : two-sided; Holm: adjusted over the four.

Contrast	Estimate	SE	p	Holm p
Culture: Korean – US, gap after rough. + harm. (harmonic tones)	−0.14	0.12	.227	.454
Training: musicians – non-musicians, gap after rough. + harm. (RE, $k = 8$)	+0.10	0.06	.121	.364
Third harmonic: observed – roughness-predicted change in the dip	+0.17	0.08	.045	.180
Roll-off: observed – roughness-predicted change in the gap, 2 vs 12 dB/oct	−0.04	0.11	.746	.746

Table 10: Differences between groups [95% bootstrap CI], SD units. Musicianship rows give the number of musicians/non-musicians. Bold: CI excludes zero (uncorrected).

	Observed gap	Gap after rough. + harm.	Observed dip	Dip after rough. + harm.
<i>Korean minus US listeners, same stimuli</i>				
Harmonic	−0.13 [−0.37, +0.10]	−0.14 [−0.39, +0.08]	−0.13 [−0.31, +0.07]	−0.14 [−0.32, +0.03]
Stretched	−0.15 [−0.36, +0.07]	−0.16 [−0.36, +0.07]	−0.03 [−0.22, +0.17]	−0.02 [−0.21, +0.18]
Compressed	−0.23 [−0.45, −0.02]	−0.12 [−0.34, +0.09]	−0.04 [−0.22, +0.17]	+0.01 [−0.19, +0.20]
<i>Musicians minus non-musicians (US)</i>				
Harmonic, 3 dB/oct (69/91)	+0.17 [−0.13, +0.44]	+0.13 [−0.19, +0.41]	+0.36 [+0.10, +0.61]	+0.18 [−0.05, +0.40]
5 equal harmonics (63/84)	+0.08 [−0.15, +0.33]	−0.10 [−0.36, +0.17]	+0.26 [+0.01, +0.53]	−0.05 [−0.28, +0.22]
5 harmonics, no 3rd (64/87)	+0.10 [−0.11, +0.30]	+0.01 [−0.20, +0.21]	+0.29 [+0.10, +0.47]	+0.27 [+0.07, +0.45]
Flute (86/102)	−0.03 [−0.31, +0.23]	−0.08 [−0.36, +0.19]	+0.06 [−0.16, +0.30]	−0.15 [−0.38, +0.10]
Guitar (86/100)	+0.30 [+0.01, +0.61]	+0.28 [0.00, +0.58]	+0.25 [−0.02, +0.52]	+0.18 [−0.09, +0.44]
Piano (73/117)	+0.38 [+0.07, +0.66]	+0.35 [+0.05, +0.64]	+0.37 [+0.12, +0.60]	+0.27 [+0.02, +0.50]
Pure tones (64/87)	+0.18 [−0.07, +0.44]	+0.15 [−0.10, +0.41]	+0.09 [−0.16, +0.32]	+0.01 [−0.24, +0.25]
Bonang (harmonic bass) (71/75)	+0.29 [−0.03, +0.61]	+0.17 [−0.15, +0.48]	+0.37 [+0.13, +0.62]	+0.19 [−0.04, +0.42]
Stretched (octave 2.1) (62/91)	0.00 [−0.26, +0.27]	+0.01 [−0.26, +0.28]	+0.04 [−0.20, +0.28]	+0.03 [−0.20, +0.27]
Compressed (octave 1.9) (68/86)	−0.21 [−0.47, +0.07]	−0.20 [−0.46, +0.08]	+0.06 [−0.17, +0.29]	+0.08 [−0.15, +0.31]

mainly a steeper decline of pleasantness with interval size, not a different ordering of consonances. Its cause is open: sharpness, which rises with interval size, is one candidate, and in an online study so is the playback equipment, which may differ between cohorts.

5.7 Chords

Scored in equal temperament, as in *incon* and in earlier analyses, including one by the author, the twelve Bowling dyads show the pattern of the dense data (Table 11): roughness reproduces −2% of the tritone–minor-sixth difference and roughness plus harmonicity 48%. In the chords, each tritone costs +0.205 rating units with only chord size controlled and +0.080 [+0.006, +0.150] after roughness and harmonicity. But the chords were played in just intonation. At the presented tuning, harmonicity rates the minor sixth (8/5) higher (1.140 against 1.070 in equal temperament), while the tritone (7/5) barely changes (0.952 against 0.943). Scored as presented, roughness plus harmonicity reproduces 93% of the dyad difference and the chord-level tritone penalty falls to +0.009 [−0.056, +0.074]. With twelve dyads the share is an overfit-prone statistic; the chord-level penalty, estimated from 117 chords containing a tritone, is the firmer result.

The Johnson-Laird chords were equal-tempered, and there a penalty remains: +0.39 [+0.08, +0.69] rating units per tritone after roughness and harmonicity (rating SD 1.10), and +0.34 [−0.01, +0.68] with familiarity added. Those chords are widely voiced and were rated by a different population, so the two sets are not directly comparable. The 20-cent tolerance reduces the Johnson-Laird penalty to +0.28 [−0.08, +0.60] (paired difference −0.11 [−0.21, −0.03]), no

Table 11: [Bowling et al. \(2018\)](#): share (%) of the tritone–minor-sixth dyad difference reproduced by each fitted model (12 dyads), and the tritone penalty in chords (rating units per tritone, chord size controlled [95% CI]; $n = 298$), with the models scored at equal-tempered (ET) or the presented just-intoned (JI) pitches.

Model	Dyad share (%)		Chord tritone penalty	
	ET	JI	ET	JI
Chord size only	–	–	+0.205	+0.205
Roughness (H&K)	–2	–2	+0.197 [+0.125, +0.268]	+0.198 [+0.125, +0.269]
Roughness (Sethares)	15	15	+0.184 [+0.115, +0.255]	+0.184 [+0.115, +0.255]
H&K + harmonicity	48	93	+0.080 [+0.006, +0.150]	+0.009 [–0.056, +0.074]
H&K + familiarity	44	44	+0.090 [+0.016, +0.161]	+0.094 [+0.020, +0.165]
H&K + harmonicity + familiarity	48	95	+0.002 [–0.070, +0.072]	–0.058 [–0.130, +0.009]
H&K + periodicity (Stolzenburg)	33	32	+0.109 [+0.037, +0.177]	+0.113 [+0.044, +0.178]

longer distinguishable from zero, and fits the just-intoned Bowling chords worse (R^2 0.739 to 0.701), so the tolerance that suits listeners depends on the tuning they hear. Rescoring the Bowling chords also changes the unique variance of harmonicity relative to interference and familiarity: with chord size controlled it rises from 0.037 to 0.068, comparable to the other two, and the full model’s R^2 rises from 0.738 to 0.775. Without a chord-size control familiarity appears to dominate, as an earlier analysis by the author reported (Supplementary Table [S12](#)).

5.8 How much the curves miss overall

The split-half ceilings range from 0.82 to 0.97 (US mean 0.93). Held out, the recommended curve captures 59% of the explainable variance in the US experiments and the composite of [Marjieh et al. \(2024\)](#) 53%. The shortfall is not at the tritone. Fitted to single experiments, the residuals share a broad shape: intervals in the middle of the range are rated above the curve and wide intervals, above all beyond the octave, below it. A linear term in interval size removes most of this (in-sample R^2 0.38 to 0.73 for harmonic tones, and a Korean mean of 0.08 to 0.87). Its slope depends on the spectrum: -0.43 SD per octave of interval for harmonic tones and -0.33 for pure tones, but -0.01 for five equal harmonics and $+0.37$ for the flute; for Korean listeners -1.03 to -0.89 . Interval size is confounded here with the pitch of the upper tone and with sharpness, the weighting of a spectrum towards high frequencies ([Fastl & Zwicker, 2007](#)). Sharpness was a component of sensory pleasantness in psychoacoustics ([Aures, 1985](#)) before consonance models narrowed to roughness and harmonicity, it lowers the consonance of chords in high registers ([Eerola & Lahdelma, 2022](#)), and for pure-tone dyads preference falls as mean frequency rises ([Kaklamani & Simserides, 2026](#)).

As an exploratory comparison, chosen after seeing the residuals, we added one smooth term with a weight shared by all experiments. Held-out US R^2 rises from 0.548 to 0.625 with interval size and to 0.658 with a sharpness proxy computed from the synthesis parameters (paired difference $+0.109$ [$+0.031$, $+0.141$]), that is from 59% to 71% of the explainable variance; widening the tolerance, by comparison, added $+0.015$. Sharpness’s edge over interval size is the part that selection may have produced. The gain does not depend on the roughness model: with Marjieh et al.’s revised roughness the US R^2 rises from 0.501 to 0.617. It is concentrated in the synthetic spectra (harmonic tones 0.26 to 0.57, stretched 0.43 to 0.84), and two spectra do not fit it: for five equal harmonics the trend is absent although sharpness rises with the interval, and the flute, whose ratings rise with interval size, loses (0.69 to 0.48). The roll-off experiment, which varied the spectral slope across the trials of the same participants, offers a check within listeners, although we looked at it before choosing sharpness. Fitted separately at each roll-off, the slope

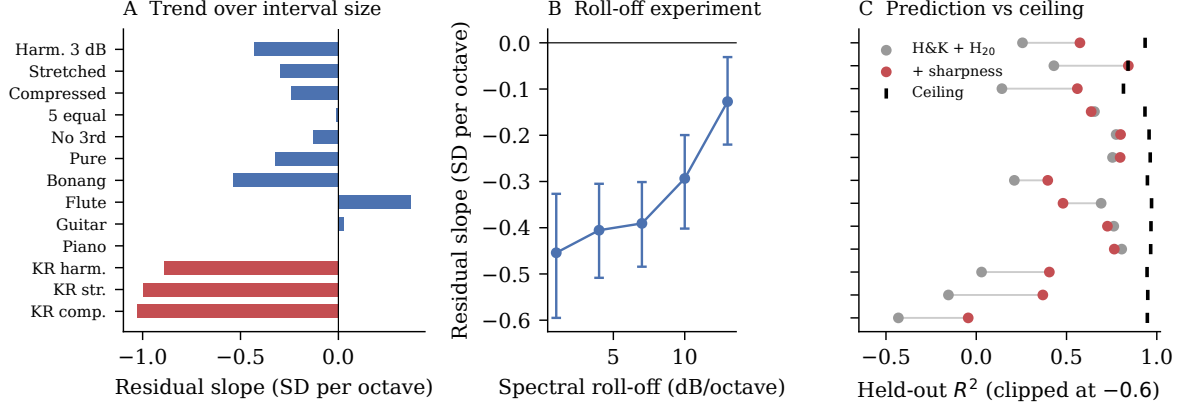


Figure 4: The smooth trend (exploratory). (A) Slope over interval size of the residual from H&K roughness plus harmonicity (20-cent tolerance), each experiment fitted alone (SD per octave of interval). (B) The same slope in the roll-off experiment at each spectral roll-off (participant-bootstrap 95% CI). (C) Held-out R^2 per experiment without and with the sharpness term, against the split-half ceiling (Korean experiments predicted with weights from all US experiments).

over interval size flattens from -0.45 $[-0.60, -0.33]$ at 1 dB/octave to -0.13 $[-0.22, -0.03]$ at 13, as sharpness predicts and interval size alone does not, but the proxy’s implied slopes cover only 50% of that flattening. Finally, for the one rating set of [Marjeh et al. \(2024\)](#) not yet analysed, 80 triads of five equal harmonics rated by 192 participants, we wrote down three predictions before reading the data: a negative sharpness weight, a better leave-one-chord-out R^2 , and a better correlation of the frozen-weight curve with the chord means. None passed (weight -0.095 $[-0.193, +0.010]$; R^2 difference $+0.005$ $[-0.009, +0.041]$; correlation difference $+0.006$ $[-0.006, +0.017]$; Supplementary Section S8), although the triads leave little room for a third term. We therefore treat the smooth trend as established and sharpness as a candidate explanation, and we do not add it to the recommended curve.

6 Discussion

6.1 What closes the gap, and the recommended curve

For harmonic tones and US listeners the minor sixth is rated $+0.26$ within-listener SD above the tritone, and a roughness curve calibrated on the rest of the interval axis predicts none of it: roughness is almost flat between the fourth and the fifth. The popular diagnosis, that the dissonance curve is an oversimplification, is correct in a precise sense. What is missing is harmonicity. The tritone’s partials fit no single harmonic series well, whereas those of the minor sixth approximate the series whose 5th and 8th harmonics are the two fundamentals. With harmonicity in the curve the tritone’s own shortfall is no longer detectable, and root ambiguity, periodicity and harmonic entropy add nothing: an interval whose partials fit no single series is the same interval whose virtual pitch is ambiguous.

The curve we recommend is $P(x) = \alpha_e + b_R R_{\text{HK}}(x) + b_H H_{20}(x)$, with R_{HK} the Hutchinson–Knopoff roughness of the dyad’s combined spectrum, H_{20} Harrison and Pearce’s harmonicity computed with a 20-cent tolerance (11-harmonic template weighted $1/i$, the same Gaussian width for spectrum and template, lower tone at C4), both smoothed with the 0.2-semitone kernel, and pooled weights $b_R = -1.65$ and $b_H = 0.66$ for the 20-cent curve (per unit of each raw model output; α_e is an experiment intercept). It has no interval-specific term. It predicts

held-out US experiments better than the published combinations we tested, but only slightly better than the same curve with the published tolerance; for just-intoned tones the published tolerance fits better. Neither the curve family nor its ingredients are new (Parncutt et al., 2019; Harrison & Pearce, 2020; Marjeh et al., 2024; Masina et al., 2022); what is new is that its tritone predictions were tested on experiments not used to fit it, against candidates fixed in advance, and on a spectrum that moves the intervals.

Three cautions apply to anyone who uses it. First, its weights are for new spectra of the kind tested here; on the experiments themselves they were refitted without the experiment being predicted, and the intercept makes the curve a statement about the shape of the profile, not its level. Second, the tolerance is a smoothing choice as much as a psychoacoustic parameter, since it interacts with the kernel that smooths ratings and model outputs alike; at a finer resolution the 20-cent curve smooths away the minor-sixth peak that listeners show (Supplementary Section S9). Third, the curve is meant for equal-tempered tones; chords presented in just intonation should be scored as presented and with the published tolerance. A smooth term over interval size would improve it more than anything tested at the tritone, but until its cause is known we leave it out.

6.2 What remains, and how to read it

What remains in dyads is the primary estimand, $+0.18\text{SD}$ $[+0.04, +0.32]$, about 0.24 points on a seven-point scale for the descriptive calibration. It is not a constant in three respects. It varies between spectra more than sampling error allows (prediction interval $[-0.18, +0.55]$), with the largest values for the proxy spectra of the sampled instruments. It is absent at 600 cents when the partials are stretched or compressed. And it is not detectably the tritone’s: after harmonicity the tritone’s residual does not differ from zero, and the minor sixth above it, like the other consonances. The exploratory decomposition does not change this. Its step is carried by two proxy spectra, and the tritone’s reduced in-tune bonus does not separate it from the other intervals once spectra are treated as a random sample; the one contrast that is clear, a smaller bonus for the major second and minor seventh, was found after seeing the profile. We would not add a tritone penalty to a dissonance model. We would add harmonicity and look for what makes consonant intervals of harmonic tones more pleasant than any current curve predicts.

Pure tones. Pure tones might seem to cut the other way: they have no upper partials to align, yet they leave one of the largest gaps. The residual profile resolves this. For pure tones the tritone is predicted about right (-0.01) and the gap is the minor sixth ($+0.18$), so pure tones give no evidence of a penalty at 600 cents in the first place. Because their partials are the fundamentals, pure tones cannot separate spectral accounts from accounts based on the frequency ratio; only a spectrum whose partials depart from the harmonic series can.

The stretched tones and the heard octave. With stretched partials the dip leaves 600 cents and appears at the stretched scale’s tritone. An earlier analysis by the author read this as compatible with a learned category defined relative to the octave the listener actually hears. That reading predicts more than the dip: a category re-centred on the heard octave should also carry the in-tune bonus to the stretched scale’s intervals. It does not ($+0.02$ $[-0.02, +0.05]$, with a detectable effect of 0.05, against $+0.13$ $[+0.10, +0.17]$ for harmonic tones). The dip follows the partials and the bonus does not, which favours a spectral account of the dip, in which partials of the two tones line up or fail to, over a re-centred learned category. This fits the finding that consonance judgements for inharmonic bell timbres follow their partials rather than the nominal

intervals (Harrison & MacConnachie, 2024). It rests on one stretched spectrum; the compressed tones show no clear dip at their own tritone.

Two readings of the minor-sixth residual. The first is that the harmonicity model is incomplete. The tuning analysis does not identify the defect: the listeners’ minor-sixth peak lies between equal temperament and just intonation, and the gap is no smaller at just-intoned positions. A harmonicity model wrong in some other detail would have to explain why the residual varies between spectra, why it is shared by the other consonant intervals, why it vanishes in the just-intoned Bowling chords and why a penalty persists in the equal-tempered Johnson-Laird chords. The second reading is learned: familiarity with the consonant intervals of tonal music, heard mostly in harmonic sounds (Lahdelma et al., 2022; McDermott et al., 2016). The in-tune bonus fits this reading, since it appears for harmonic tones and not at either scale of the stretched and compressed ones. But the targeted tests do not support its usual corollaries once corrected for multiplicity: neither training (compare McDermott et al., 2010; Popescu et al., 2019) nor culture changes the residual detectably. These two readings are those of the current debate, between psychoacoustic accounts in which harmonicity is a perceptual mechanism (Harrison & Pearce, 2020; Marjeh et al., 2024) and the view that harmonicity partly stands in for exposure (Lahdelma et al., 2022; Milne et al., 2023). Our data locate the residual at consonant intervals of harmonic tones; they cannot tell which reading is right.

The chords sharpen the contrast without resolving it. The just-intoned Bowling chords leave no tritone penalty after roughness and harmonicity, and the equal-tempered Johnson-Laird chords leave one, which the 20-cent tolerance reduces to a value whose interval includes zero (+0.28 [−0.08, +0.60]). This is what an incomplete harmonicity model would produce if it under-credits equal-tempered consonances, and it is also what a learned preference for familiar equal-tempered intervals would produce. The two chord sets differ in voicing and population as well as tuning, and in chords the tritone penalty is estimated from a count of tritones, so the chords do not show that the tritone is special either; they show that tuning must be scored as presented.

What would decide. Three experiments would separate the readings. A replication of the stretched-spectrum result with a second stretch factor and a second cohort would test whether the dip follows the partials generally, and a dense sampling of the minor-sixth and major-third regions at a resolution finer than the 14 cents that separate the two tunings, with enough listeners per spectrum, would locate the peaks well enough to decide whether the harmonicity model misplaces them. A comparison of listeners with little exposure to equal-tempered harmonic music, rating the same dense dyads, would test the learned reading directly; the Korean cohort here is too small and too different in recruitment to do so. Finally, the dip and the gap should be measured in musical context, where the tritone’s reputation was formed: the isolated dyads here say how a tritone sounds, not how it functions.

6.3 What dissonance curves can say

Roughness curves computed from measured instrument spectra do not settle the ordering of the tritone and the minor sixth. Across eight instruments at C4 and four kernels, the instrument accounts for 55% of the variance of the tritone/minor-sixth roughness difference, the kernel for 8% and their interaction for 36%; the rarely reported shape constants of the H&K kernel move it by a median 13.9 pp, and one clarinet spectrum reverses the listeners’ ordering under H&K in two libraries (Supplementary Section S11). Combined with the finding that listeners do not

follow roughness in this region even for idealised spectra, roughness curves should not be used to argue about the tritone either way. We recommend that analyses report the kernel and its constants, the partial-merging and peak-acceptance rules, the recording and register, the tuning at which the stimuli were presented, and whether the conclusion holds for a second recording; and that any new curve be judged on held-out spectra and against the noise ceiling, not against the previous curve.

7 Limitations

The dense data are online pleasantness ratings of synthetic or resynthesised dyads heard in isolation; the tritone’s reputation concerns musical context, which these data do not reach. For the flute, guitar and piano the spectra are proxies from one analysed tone, and the largest residuals, and the step of the exploratory decomposition, are found there. Models were evaluated with the lower tone at C4, whereas it was sampled from G3 to F4; averaging roughness over that range changes the held-out R^2 of the recommended curve by -0.001 $[-0.001, 0.000]$ and its dip by $+0.001$ $[+0.001, +0.001]$. Six harmonic experiments are few for estimating heterogeneity, which is large; Hartung–Knapp intervals are wider than the DerSimonian–Laird intervals reported earlier, and 22 of 326 pooled estimates change whether they exclude zero. Experiments are treated as independent although some participants may have taken part in more than one. The primary estimand was declared in this version, after the author’s earlier analyses, although the question and the curve were fixed before; the screen’s candidate list was fixed in advance but not publicly registered. The in-tune bonus, the step, the residual profile, the stretched-scale bonus, the tuning analysis, the sharpness term and the Korean held-out fits are exploratory. The four targeted tests are corrected for multiplicity; the many other contrasts are not. The stretched-scale results rest on one spectrum and one cohort, and the test of the heard-octave account assumes that a re-centred category would carry the bonus as well as the dip. The tuning analysis could not separate equal temperament from just intonation: the peak’s interval spans both, and equivalence was not shown (Supplementary Section S9). The Korean cohort is small, differs from the US cohort in recruitment and playback equipment as well as culture, and its profiles are poorly fitted. The musicianship split is observational with a coarse two-year threshold. Roughness here is a model quantity, not a measured percept (McBride, 2025). The Bowling analysis rests on twelve dyads and 298 chords from 30 listeners; the chord bootstraps resample chords, not raters; and the Johnson-Laird chords differ in voicing, tuning and population. The sharpness proxy is computed from synthesis parameters, was chosen after seeing the residuals and failed its pre-specified (not publicly registered) triad test. No new listener data were collected.

8 Conclusion

Listeners rate the tritone below the minor sixth, and a roughness curve explains none of that. A curve with harmonicity added, tolerating more mistuning than the published model, predicts held-out experiments slightly better than published curves and leaves a gap of $+0.18$ within-listener SD $[+0.04, +0.32]$ that varies between spectra and is unexplained. That gap sits at the minor sixth, which listeners rate above the curve as they do the other consonances; the tritone’s own residual does not differ detectably from zero. With stretched partials the tritone’s dip follows the partials, and none of the tests of training, culture, third harmonic or roll-off survives correction. The larger failure of dissonance curves is a smooth trend over interval size, which leaves them far below the reliability of the ratings. In dyads, harmonicity accounts for

the shortfall, and what remains is not detectably the tritone's.

Data availability

No new data were collected; the study reanalyses published data. The dense dyad ratings and timbre parameters are from Harrison (2023) (<https://gitlab.com/pmcharrison/timbre-and-consonance-paper>), the chord ratings from Digital and Cognitive Musicology Lab, EPFL (2025) (<https://github.com/DCMLab/consonance>) and Harrison (2019) (<https://github.com/pmcharrison/inconData>), and the instrument recordings from Philharmonia Orchestra (n.d.) (<https://philharmonia.co.uk/resources/sound-samples/>) and University of Iowa Electronic Music Studios (n.d.) (<https://theremin.music.uiowa.edu/MIS.html>). The code repository provides scripts that fetch these data rather than copies; the partials extracted from the recordings are included as derived data.

Code availability

All analysis code, intermediate results and the manuscript source are available at <https://github.com/boenchen1112/tritone-harmonicity-dyads> (release v1.0-submission; archived at [ZENODO DOI]). Every number in the text and tables is generated from JSON result files by `analysis/make_tables.py`, and every figure by `analysis/make_figures.py`; the exceptions are design constants and values quoted from the literature.

Ethics statement

This is a secondary analysis of published, anonymised data. No new data were collected from human participants, and no ethical approval was required.

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Competing interests

The author declares no competing interests.

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